Simple Steps - Power BI Lab (very easy version)

How to use: do **Part 0** first (always). Then find your practical. Do the steps one by one. Copy the same numbered steps on paper as the Procedure. Words in **bold** are buttons or menu names you click. Lines in grey boxes are text you type or paste.

PART 0 - Do this first for ANY practical

Goal: get the sales data into Power BI.

1. Open **Power BI Desktop**. Click **Blank report**.
2. Click **Home** (top menu), then **Get data**, then **Text/CSV**.
3. Choose the file your teacher gave. Click **Open**.
4. Click **Transform Data**. (A new window opens. This is where you clean data.)
5. Look at each column title. Click the small icon on the left of the title and set the type: **OrderDate = Date**, **Quantity = Whole Number**, **Revenue, Cost, UnitPrice = Fixed Decimal Number**.
6. On the right side, click the **Name** box. Delete the name and type **Fact_Sales**.
7. Click **Close & Apply** (top left). The window closes.
8. Click the **table icon** on the left side. You will see the rows. Check there are 6 rows.

If the teacher gives NO file: in step 2 instead click Home > Transform data, then Get data > Blank query > Advanced editor, paste the data code from the README (Block A) and click Done.

How to make a calculation (called a Measure)

1. Click **Home** then **New measure**.
2. Type the formula. Press **Enter**.
3. Do ONE formula at a time.

How to make a chart

1. Click an empty place on the page.
2. On the right, in **Visualizations**, click the chart picture you want (hover to see the name).
3. On the right, in **Data**, **drag** the field names into the chart boxes (X-axis, Y-axis, etc.).

PRACTICAL 1 - Import data

1. Do all 8 steps of Part 0.

2. Look at the 3 icons on the left: top = **Report**, middle = **Table**, bottom = **Model**.

3. Excel file: **Home > Get data > Excel workbook**, choose the file, tick the sheet, click **Load**.

4. Database: **Home > Get data > SQL Server**, type server and database name. (If no database is given, write 'database not available'.)

5. File > Save as, name it **Northwind_Lab**.

Write on paper (Output / Observation / Result):
Output: the 6-row table. Total quantity 32, total revenue 171,600.
Observation: data has 6 orders and total revenue 171,600.
Result: data was imported correctly.

PRACTICAL 2 - Clean the data

Use the customer file. Load it like Part 0 (name it customer_raw).

1. Click **Home > Transform data** to open the cleaning window.

2. Click on the **Transform** tab. Select all columns (click first title, then Shift+click last). Click **Format > Trim**. (Removes extra spaces.)

3. Click **Home > Remove Rows > Remove Blank Rows**. (Empty row goes away.)

4. Click the **CustomerKey** title. Click **Remove Rows > Remove Duplicates**. (Same customer twice becomes once.)

5. Click the small arrow on **CustomerName**. Untick **null** and **(blank)**. Click OK.

6. Click the **City** title. Click **Transform > Format > Capitalize Each Word**. (mumbai becomes Mumbai.)

7. Click **Close & Apply**.

Write on paper (Output / Observation / Result):
Output: rows go 7 -> 6 -> 5 -> 4.
Observation: blank row, duplicate and missing-name rows were removed, leaving 4 customers.
Result: customer data is clean.

PRACTICAL 3 - Star schema (link the tables)

Idea: one big sales table in the middle, small tables (product, region, date) around it, joined by lines.

1. Do Part 0 first. Click **Home > Transform data**.
2. On the left, right-click **Fact_Sales** and click **Reference**. Name the new one **Dim_Product**.
3. In Dim_Product click **Home > Choose Columns**. Tick only ProductKey, ProductName, Category. Click OK. Then click ProductKey title > **Remove Rows > Remove Duplicates**.
4. Do steps 2 and 3 again with the name **Dim_Region**. Keep only RegionKey and RegionName.
5. Click **Close & Apply**.
6. Click **Modeling > New table**. Paste this and press Enter:

```
Dim_Date = CALENDAR(DATE(2024,1,1), DATE(2024,12,31))
```

7. Click the **Model** icon (3rd on the left). You see boxes for each table.
8. Drag **ProductKey** from Dim_Product and drop it on **ProductKey** in Fact_Sales. A line appears. Do the same for **RegionKey**. Drag **Date** from Dim_Date onto **OrderDate** in Fact_Sales.
9. Go back to the **Report** icon. Add a **Table** chart. Drag RegionName (from Dim_Region) and Revenue (from Fact_Sales) into it.

Write on paper (Output / Observation / Result):
Output: draw Fact_Sales in the middle and 3 boxes around it with lines. Table: North 119,600, East 28,500, South 16,000, West 7,500, Total 171,600.
Observation: region totals add up to 171,600, so the links work.
Result: star schema was created.

PRACTICAL 4 - Basic charts

1. Do Part 0 first.
2. Make a measure (Home > New measure):

```
Total Sales = SUM(Fact_Sales[Revenue])
```

3. Make another measure:

```
Total Quantity = SUM(Fact_Sales[Quantity])
```

4. **Column chart:** click the 'Clustered column chart' picture. Drag **RegionName** to X-axis and **Total Sales** to Y-axis.
5. **Line chart:** click the 'Line chart' picture. Drag **OrderDate** to X-axis and **Total Sales** to Y-axis. (In the X-axis box click the small arrow next to OrderDate and choose OrderDate, not 'Date Hierarchy'.)
6. **Donut chart:** click the 'Donut chart' picture. Drag **Category** to Legend and **Total Sales** to Values.
7. **Treemap:** click the 'Treemap' picture. Drag **Category** to Group, **ProductName** to Details and **Total Sales** to Values.
8. Put all 4 charts on one page. **File > Save**.

Write on paper (Output / Observation / Result):
Output: draw the 4 charts. Numbers: North 119,600, East 28,500, South 16,000, West 7,500. Electronics 148,100 (86.31%), Apparel 23,500 (13.69%).
Observation: North has the highest sales and Electronics gives most of the revenue.
Result: column, line, donut and treemap charts were created.

PRACTICAL 5 - Calculations (DAX)

1. Do Part 0 first.

2. Click the table **Fact_Sales** on the right. Click **Table tools > New column**. Type:

```
Profit = Fact_Sales[Revenue] - Fact_Sales[Cost]
```

3. New column again:

```
Order Size = IF(Fact_Sales[Revenue] > 50000, "Large", "Standard")
```

4. Make these measures one by one (Home > New measure):

```
Total Sales = SUM(Fact_Sales[Revenue])
Total Profit = SUM(Fact_Sales[Profit])
Total Orders = DISTINCTCOUNT(Fact_Sales[OrderID])
AOV = DIVIDE([Total Sales], [Total Orders])
Profit Margin = DIVIDE([Total Profit], [Total Sales])
```

5. (Type each line in its own New measure box. Not all together.)

6. Add a **Table** chart. Drag OrderID, Revenue, Cost, Profit, Order Size into it.

7. Add 4 **Card** charts. Put Total Sales, Total Profit, AOV, Profit Margin in them.

8. Add a **Slicer** chart with RegionName. Click North and see the cards change. Click it again to clear.

Write on paper (Output / Observation / Result):
Output: Total Sales 171,600, Profit 43,100, AOV 28,600, Margin 25.12%. With North: Sales 119,600, Profit 28,100.
Observation: only order S101 is Large. Measures change when a filter is used.
Result: columns and measures were created.

PRACTICAL 6 - Slicers and drill through

1. Do Part 0 and make the measures from Practical 4 and 5.

2. Add 3 **Slicer** charts: RegionName, Category, OrderDate.

3. Click Electronics in the Category slicer. Look at the numbers. Click again to clear. Do the same for North.

4. Click a chart. Look at the **Filters** pane on the right. Drag Category into 'Filters on this page' and choose 'is not blank'.

5. Click the **+** at the bottom to add a new page. Double-click its name and type **Product Details**.

6. On this page, find the **Drill through** box in the right pane. Drag **ProductName** into it.

7. Add a **Table** chart with ProductName, OrderID, Quantity, Revenue, Profit.

8. Go to the first page. Add a chart that has ProductName. **Right-click Laptop > Drill through > Product Details**.

Write on paper (Output / Observation / Result):
Output: Electronics = 3 orders, 148,100. North = 2 orders, 119,600. Laptop page shows order S101, 110,000.
Observation: slicers change numbers; drill through shows details of one product.
Result: slicers and drill through work.

PRACTICAL 7 - KPI and scorecard

Needs the monthly file (MonthStart, ActualRevenue, TargetRevenue). Load it like Part 0, name it monthly_data.

1. Load the monthly data (Part 0 steps, but name it monthly_data).

2. Make measures one by one:

```
Actual Revenue = SUM(monthly_data[ActualRevenue])
Target Revenue = SUM(monthly_data[TargetRevenue])
Attainment = DIVIDE([Actual Revenue], [Target Revenue])
```

3. Add a **Column chart**: MonthStart on X-axis, Actual Revenue on Y-axis.

4. With the chart selected click the **KPI** picture. Put Actual Revenue in Value, MonthStart in Trend axis, Target Revenue in Target.

5. Add a **Table** with MonthStart, Actual Revenue, Target Revenue, Attainment.

6. For colours: in the table box click the arrow next to **Attainment** > **Conditional formatting > Background color**. Make rules: 1 or more = green, 0.9 to less than 1 = amber, less than 0.9 = red.

Write on paper (Output / Observation / Result):
Output: Jan 105% green, Feb 90.48% amber, Mar 113.33% green, Apr 84.29% red.
Observation: April is below target, March is above.
Result: KPI and colour scorecard were created.

PRACTICAL 8 - Sales dashboard

1. Do Part 0 and the measures from Practical 4 and 5.

2. Add 4 **Card** charts on top: Total Sales, Total Orders, Total Profit, AOV.

3. Add a **Bar chart**: RegionName and Total Sales.

4. Add a **Column chart**: Category and Total Sales.

5. Add a **Line chart**: OrderDate and Total Sales.

6. Add 3 **Slicers**: RegionName, Category, OrderDate.

7. Click each slicer and check the numbers change. Clear them.

8. **Home > Publish**. (This needs a work account. If it will not open, write the step and say account not available.)

Write on paper (Output / Observation / Result):
Output: sketch the page. Cards: 171,600 / 6 / 43,100 / 28,600.
Observation: charts change when slicers are used.
Result: sales dashboard was created.

PRACTICAL 9 - HR dashboard

Needs the HR file (EmployeeID, Department, JoiningDate, ExitDate, Status, Salary). Load like Part 0, name it hr_dashboard_data.

1. Load the HR data. Make sure JoiningDate and ExitDate are type **Date**.

2. Make measures one by one:

```
Total Employees = DISTINCTCOUNT(hr_dashboard_data[EmployeeID])
Exited Employees = CALCULATE([Total Employees], hr_dashboard_data[Status] = "Exited")
Exit Percent = DIVIDE([Exited Employees], [Total Employees])
```

3. Add 3 **Card** charts: Total Employees, Exited Employees, Exit Percent.

4. Add a **Column chart**: Department on X-axis, Exit Percent on Y-axis.

5. Add a **Table** with EmployeeID, Department, JoiningDate, ExitDate, Status.

6. New page. Drag **Department** into the **Drill through** box. Add the same table there.

Write on paper (Output / Observation / Result):
Output: 5 employees, 2 exited, 40%. HR 100%, IT 100%, Sales 0%, Finance 0%.
Observation: only 5 people, so 40% is not a real company figure.
Result: HR dashboard was created.

PRACTICAL 10 - Full project

1. Do Practicals 1, 3, 4, 5 and 7 first (data, links, measures, charts).

2. Page 1: 4 cards and a line chart. Name the page **Executive Summary**.

3. Page 2: column chart of RegionName and Total Sales. Name it **Regional Analysis**.

4. Page 3: table with ProductName, Revenue, Cost, Total Profit. Name it **Product Analysis**.

5. Page 4: the KPI and table from Practical 7. Name it **Performance Scorecard**.

6. Copy a slicer to every page: click **View > Sync slicers** and tick the pages.

7. Home > Publish. (If no account, write the step.)

Write on paper (Output / Observation / Result):
Output: sketch the 4 pages. Insights: North is 69.70% of sales, Electronics 86.31%.
Observation: sales are concentrated in North and Electronics; data is only 2 months.
Result: four-page report was created.